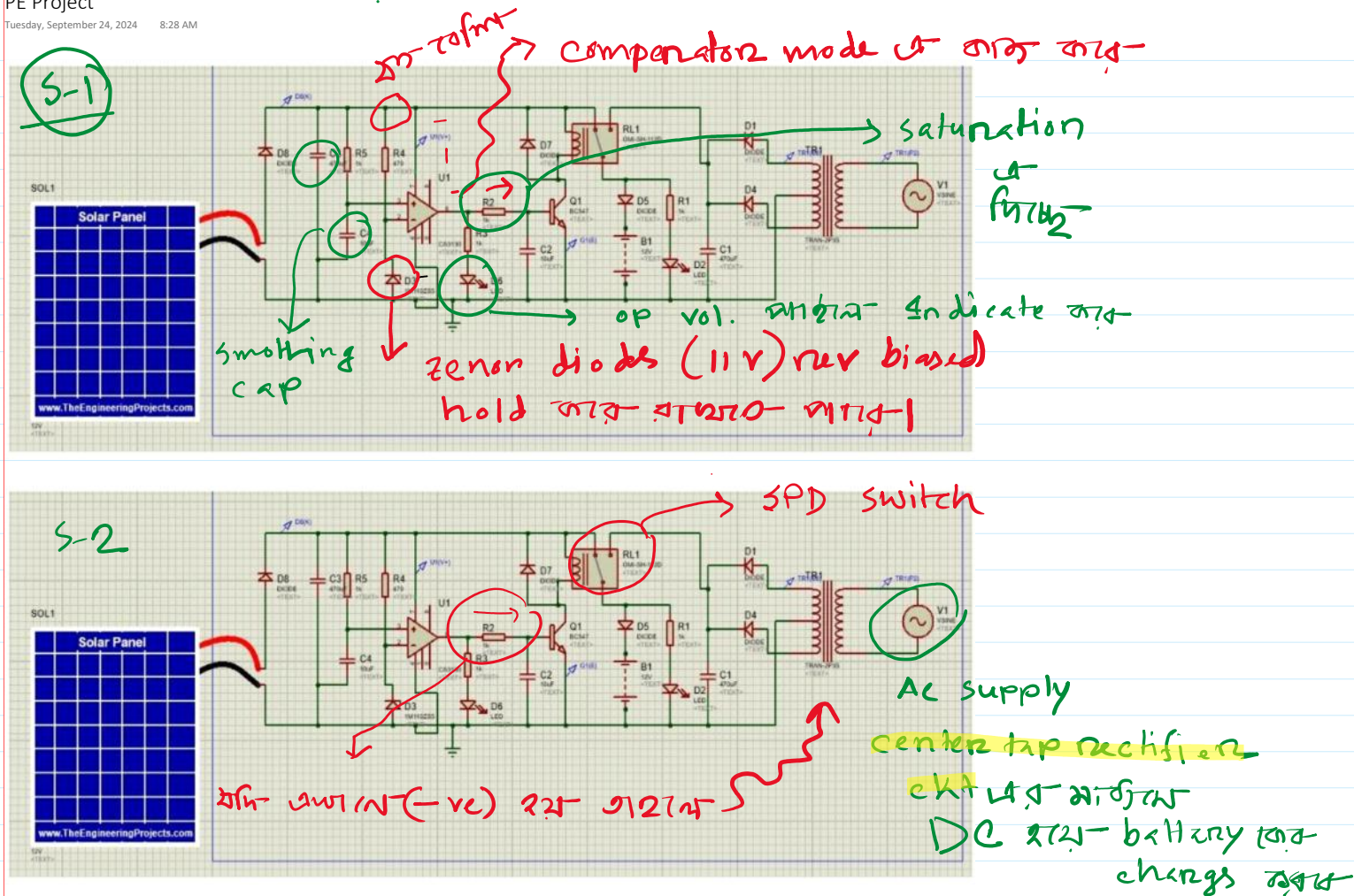




The working process of the Hybrid Inverter with Solar and AC Main Battery Charger circuit involves two key circuits: the charging circuit and the inverter circuit. Here's a detailed explanation of how each of these circuits functions:

1. Charging Circuit:

1. The charging circuit's primary function is to charge the battery, and it does so using two power sources: solar energy and AC mains.
2. Solar Charging:
3. When the output from the solar panel is 12 Volts or higher, the battery is charged using solar power.
4. The solar panel outputs 18 Volts under bright sunlight. A Zener diode is used to step down this voltage to around 11 Volts, which is fed to a comparator's inverting terminal.
5. The comparator compares the voltages, and if the non-inverting input gets a higher voltage than the inverting terminal, the output becomes high. This high output triggers a transistor to turn on, which then energizes a relay.
6. When the relay is energized, the battery starts charging via the solar panel. A green LED indicates the battery is charging from the solar power.

AC Mains Charging (Fallback):

1. If the solar panel's output falls below 12 Volts (due to cloudy weather or night), the system automatically switches to the AC mains to charge the battery.
2. This switching is done by a Single Pole Double Throw (SPDT) relay. When the solar voltage is

insufficient, the relay de-energizes and the battery starts charging from an AC mains supply through a step-down transformer and a rectifier circuit (which converts AC to DC).

3. The changeover between solar and AC mains charging is seamless, ensuring the battery always remains charged.

2. Inverter Circuit:

1. The inverter circuit converts the DC power stored in the battery into AC power to be used by appliances.
2. The SG3525A IC is used to generate PWM (Pulse Width Modulation) signals, which are required to control the switching of MOSFETs.
3. The PWM signals from the SG3525A are used to switch the MOSFETs on and off. MOSFETs are high-speed switches that are responsible for converting DC power from the battery into a modulated sine wave (AC).
4. Output A from the SG3525A controls the first three MOSFETs, and Output B controls the other three.
5. This switching arrangement creates a sine wave output, which can be used to power AC devices like bulbs, fans, or other household appliances.

Key Points of the Circuit:

- The relay is the central component that ensures automatic switching between solar power and AC mains for charging the battery.
- The inverter circuit ensures that even when solar power or AC mains are unavailable, the system can still provide uninterrupted AC power using the battery.
- The system optimizes solar energy use and ensures a continuous power supply even during power outages.
- In summary, the charging circuit uses solar energy when available and switches to AC mains when necessary. The inverter circuit converts the stored DC power from the battery into AC power for household use, ensuring a stable, uninterrupted power supply.